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Synthesis of Energetics

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Chapter 1 Sufficiency

1.1 Context and Motivation

Global climate targets are under pressure. The 6th IPCC report (Working Group 3) confirms with high confidence that the world is behind on addressing climate change. Most 2°C scenarios (two thirds of them) rely on negative carbon technologies such as Direct Air Capture (DAC) or Bioenergy with Carbon Capture and Storage (BECCS), yet these technologies have been studied for over 50 years without becoming economically viable.

The remaining global carbon budgets are tight. Updated 2023 estimates put the budget at approximately 260 $GtCO_2$ for a 50% chance of limiting warming to 1.5°C, and around 950 $GtCO_2$ for a 67% chance of staying below 2°C. For context, proven fossil fuel reserves embed some 3 500 $GtCO_2$ of potential emissions — far beyond any safe budget.

1.2 The Sufficiency–Efficiency–Renewables Framework

The presentation applies the *negaWatt* approach, built on three complementary levers formalized through the Kaya identity:

- **Energy sufficiency** — reducing the level of a need through behavioural and societal change (e.g. smaller dwellings, less car use, dietary shifts). Introduced in IPCC AR6 WG3 as a policy category in its own right.
- **Energy efficiency** — reducing the energy required to deliver a given service through better technology (heat pumps, insulation, electric vehicles).
- **Renewable energy** — decarbonizing the remaining energy supply.

A key conceptual distinction is made between sufficiency and efficiency: efficiency improvements can be cancelled out by growing consumption (the rebound effect). German residential data illustrate this clearly — gains in energy intensity per m^2 have been largely offset by rising living space per capita.

1.3 Sectoral Demand Scenarios

The negaWatt Belgium scenario covers agriculture, residential, tertiary, mobility, freight, industry, non-energy-demand (NED) industry, and aviation bunkers.

1.3.1 Buildings

Residential space heating (65 TWh in 2019) is projected to fall by 71% to 19 TWh in 2050, driven by:

- Temperature setpoint reduction: 21 °C → 19 °C (−12%)
- Building envelope renovation at 3%/year from 2030 (−28%)
- Heating technology switch (18% → 82% heat pumps and district heating) (−20%)
- Improved system efficiency, average COP 2.5 → 3.5 (−10%)

Combined with the tertiary sector, total building final energy consumption falls by 48% between 2019 and 2050, with natural gas and liquid fuels being largely replaced by ambient heat and electricity.

1.3.2 Transport

Individual car mobility (56 TWh in 2019) is reduced by 93% to just 4 TWh by 2050 through a combination of sufficiency and efficiency measures:

- Distance reduction (−10%) via teleworking and urban densification
- Modal shift: car share 80% → 57% (−14%)
- Car occupancy: 1.25 → 2 persons per vehicle (−18%)
- Near-full fleet electrification, 99.5% by 2050 (−31%)
- Vehicle energy efficiency gains: 0.22 → 0.13 kWh/km (−19%)

The number of cars on Belgian roads would fall from 6 million to 2 million. Domestic transport final energy consumption declines by 83% overall.

1.3.3 Industry

Key industrial transformations include:

- **Steel**: −15% volume reduction; fuel switch from coke to hydrogen (60% direct reduction) and electric arc furnaces (40%); energy intensity 3 950 → 2 300 MWh/kt.
- **Cement**: −25% volume (more timber construction); energy intensity 904 → 650 MWh/kt; fuel switch coal → biomass.
- **Chemicals**: −20% demand; process electrification and fuel switching to renewables.

Total industrial final energy consumption falls by 34%.

1.3.4 Overall Final Energy Consumption

Belgium's total final energy consumption decreases from **490 TWh in 2019** to **246 TWh in 2050**, a reduction of **50%**, broken down as: buildings −49%, transport −76%, industry −34%.

1.4 Energy System Modelling: PyPSA-Belgium

The supply side is modelled using the open-source PyPSA-Eur framework adapted for Belgium. The model features:

- One node per country, covering five countries to represent cross-border interconnections
- Optimization runs for 2030, 2040, and 2050 to map the full transition pathway
- Hourly time resolution to capture renewable variability and flexibility requirements
- Full sector coupling: electricity, gas, hydrogen, heating, and industry

Sufficiency demand assumptions from the negaWatt Belgium scenario are fed directly as inputs to the supply-side optimization.

1.5 Two Scenarios Compared

	Reference (Ref)	**Sufficiency (nW BE)**
Emission target (Belgium, 2050)	−95%	−95%
Target (5 countries, 2050)	Net zero	Net zero
Demand	Current trends	Sufficient demand
Carbon Capture and Storage (CCS)	Allowed	Not allowed
Carbon Capture and Utilization (CCU)	Allowed	Allowed
Direct Air Capture (DAC)	Allowed	Not allowed
Building renovation rate	1%/year	3%/year
Industry	Continued trends	Increased circularity

1.6 Key Results

1.6.1 Cost-Effectiveness

The negaWatt Belgium scenario reduces total average annual system costs (CAPEX + OPEX) by **33% compared to the reference scenario** and by 19% compared to 2019 levels. Savings arise from reduced fuel import costs, avoided infrastructure investment, and lower operating costs due to a smaller overall energy system.

1.6.2 Energy Independence

The sufficiency scenario doubles Belgium's share of local energy production (from approximately 50% to 62%) and reduces energy imports by a factor of four. This significantly improves strategic autonomy, especially in combination with circular economy policies that limit dependence on critical raw materials.

1.6.3 Renewable Deployment

Sufficiency substantially reduces the renewable capacity required:

- **Solar PV**: 48 GW (reference) vs. 48 GW (sufficiency), but annual installation rate drops from 3 000 MW/year to 1 600 MW/year.
- **Onshore wind**: 14 GW (reference) vs. 7.4 GW (sufficiency); annual additions 440 MW vs. 160 MW.
- **Offshore wind**: 8 GW in both scenarios.

1.6.4 Hydrogen

Hydrogen demand in 2050 reaches 82 TWh in the sufficiency scenario, which is **43% less** than the 144 TWh needed in the reference case. Hydrogen is reserved for hard-to-decarbonize uses: industry, long-distance transport, synthetic fuel production, and non-energy applications. Most supply (84%) is imported; local electrolysis provides 13 TWh.

1.6.5 Cumulative Emissions

The sufficiency scenario achieves the −95% emission target for Belgium by 2050 without relying on CCS or DAC, using instead biomass-based carbon cycles, land use, and forestry (LULUCF) sinks. The reference scenario reaches the same 2050 target but relies heavily on large-scale carbon removal, resulting in significantly higher cumulative emissions over the transition period.

1.7 Sensitivity Analyses

1.7.1 Additional Offshore Wind

Expanding offshore capacity by an additional 16 GW (to 24 GW total by 2050, consistent with the Esbjerg and Ostend declarations) reduces onshore wind needs by 35% and increases power-to-gas capacity by 66%. Electricity and hydrogen imports also decrease. Core messages on cost-effectiveness and feasibility remain unchanged.

1.7.2 Nuclear Energy Costs

At the default cost assumption of 8 594 EUR/kW, new nuclear is not selected by the optimizer. Nuclear begins to appear in the solution only at approximately 4 500 EUR/kW — a dramatic cost reduction only achievable with small modular reactors (SMRs) under optimistic assumptions. At that price point, around 10 GW of nuclear capacity would be deployed, substituting for some onshore wind, solar PV, and reducing transmission needs.

1.8 Key Takeaways

1. **Cheapest scenario**: The negaWatt Belgium pathway is *more* cost-effective than a technology-only reference scenario, reducing system costs by 33%.
2. **Strategic independence**: Consuming less energy directly increases Belgium's energy autonomy and reduces exposure to geopolitical energy risks.
3. **Realistic scenario**: The sufficiency scenario avoids CCS and DAC, technologies that face major industrial-scale deployment challenges. Renewable deployment remains ambitious but stays well below estimated technical potentials.
4. **Fair scenario**: The scenario prioritizes collective solutions benefiting the most vulnerable citizens, treats affordable energy access as a fundamental right, and frees up global resources for the Global South by reducing Belgium's overall footprint.

The full model and data are openly available at:

- `https://github.com/IntSusEnergySystems/PyPSA-Belgium`
- `https://www.negawatt.be/scenario/`

Chapter 2 Energy market

Energy Markets — Executive Synthesis

Based on irex Consulting's lecture for UCLouvain, February 2026

2.1 Market Structure & Key Actors

The energy market involves several distinct layers of actors operating across the **energy supply chain** and the **physical infrastructure**:

- **Producers** — Generate electricity or gas (e.g. ENGIE, Luminus, TotalEnergies, Aspiravi). They dispatch power plants and inject energy into the grid.
- **Energy Markets** — Centralised trading platforms (e.g. EEX, EPEX SPOT, Nord Pool, ICE) where energy is bought and sold.
- **Balance Responsible Parties (BRPs)** — Entities responsible for balancing their portfolio of supply and demand on the market.
- **Suppliers** — Commercial intermediaries (e.g. ENGIE, Luminus, Lampiris, Eneco, Mega) who purchase energy and provide it to end consumers.
- **Transmission System Operator (TSO)** — Manages high-voltage transport of energy across the network.
- **Distribution Grid Operator (DGO)** — Distributes energy at lower voltage to end consumers.
- **Consumers** — Residential and industrial end-users.
- **Regulator** — Oversees market fairness and compliance (e.g. CREG, VREG, CWaPE, Brugel in Belgium).

2.2 Production Mix — Belgium

Historical Trends (2010–2025)

Belgium's electricity production mix has undergone significant transformation:

- **Nuclear** has declined from 55% (2010) to ≈34% (2025), but remains a dominant source.
- **Coal** has been phased out between 2010 and 2020 (from 7% to 0%).
- **Wind and Solar** have grown rapidly, collectively reaching ≈32% in 2025.
- **Gas (CCGT)** plays a flexible backup role (≈19% in 2025).

2026 Policy Update: Nuclear Is Back

The new Belgian government has repealed the nuclear phase-out law. Key developments:

- Long-term operation (LTO) extensions of **Doel 4** and **Tihange 3** are being negotiated until 2045/2055.
- A **50/50 public-private joint venture** between the Belgian State and ENGIE will manage the 2 GW LTO capacity.
- Nuclear is re-confirmed as a long-term pillar of the Belgian energy mix alongside renewables.

2.3 Green Electricity Labels

Three mechanisms certify electricity as "green" in Belgium:

1. **Green Certificates (GsC)** — Tax/subsidy mechanism for green producers; an obligation for suppliers, but not a consumer-facing label.
2. **Guarantees of Origin (GoO)** — Tracks renewable energy production; allows suppliers to market supplied energy as green. Currently at yearly granularity, with initiatives exploring finer time resolution.
3. **Power Purchase Agreements (PPAs)** — Direct physical purchase of electricity from producer to consumer.

2.4 Gas Supply — Belgium

2018 vs. 2025 Comparison

	2018	**2025**
Total Import	343 TWh	428 TWh
Domestic Consumption	187 TWh	150 TWh
Total Export	156 TWh	278 TWh
Main import sources (2025):		
Norway	48%	38%
LNG ships	8%	34%
Netherlands	32%	(net exporter)
UK	4%	13%
France	–	15%
Main export destinations (2025):		
Germany	–	50%
Netherlands	–	14%
France	44%	–

The key shift is the **East/West balance disruption** caused by the Russia–Ukraine conflict: Belgium switched from exporting to France to becoming a major transit hub exporting to Germany and the Netherlands, while LNG imports surged dramatically.

2.5 Energy Price Formation

The Merit Order

Electricity prices are set by the **marginal technology** needed to satisfy demand at any given hour. Technologies are dispatched in order of increasing marginal cost:

$$\text{Wind/Solar} \rightarrow \text{Nuclear} \rightarrow \text{CCGT} \rightarrow \text{Oil}$$

- **Normal case**: CCGT sets the marginal price (≈60 €/MWh).
- **High RES production**: Renewables or nuclear set a low marginal price.
- **Low RES & high demand**: Oil or peaking units set a very high price (>130 €/MWh).

The Clean Spark Spread (CSS)

When gas-fired plants (CCGT) are marginal, the electricity price is driven by gas and CO_2 costs:

$$\text{CSS} = P_{\text{elec}} - 2 \times P_{\text{gas}} - 0.411 \times P_{\text{CO}_2}$$

As of early 2026, with gas at ≈26 €/MWh and EUAs at ≈80 €/tCO_2, a CCGT requires power prices above **85 €/MWh** to be profitable. The Belgian forward price (CAL 27 BASE) stood at 76.27 €/MWh, implying CCGTs are *out of the money* (CSS ≈ -9 €/MWh).

Price Evolution (2021–2025)

- **2021**: Rising prices driven by post-COVID industrial restart, low wind, hot summer, and low French nuclear availability.

- **2022**: All-time high prices following Russia's invasion of Ukraine and supply chain disruption. Spot prices peaked at ≈450 €/MWh.
- **2023–2025**: Significant price correction due to supply chain reorganisation, good weather, and high gas storage levels. Prices remain above pre-COVID levels.

2.6 Supply Contracts

Energy suppliers offer contracts spanning a spectrum from fully fixed to fully variable:

	Fixed Price	**Variable Price**
Mechanism	Single €/MWh price locked at signing	Price varies each hour with spot market
Price driver	Futures market	Day-Ahead SPOT
Hedging strategy	Supplier buys in advance	Supplier buys on SPOT DA
Customer risk	Low	High
Supplier risk	High	Low

Index and *Click* products sit between the two extremes.

2.7 Supplier Differentiation

Since electricity and gas are commodities (identical products), suppliers cannot differentiate on product quality. Competitive levers are:

- **Price** — With razor-thin gross margins (≈5%), cost efficiency is paramount.
- **Customer service** — Operational excellence in billing, switching, and support.
- **Marketing** — Green image, local identity, brand trust.
- **Extra services** — eMobility, flexibility management, apps, maintenance.

2.8 Energy Market Evolutions

Cannibalization Effect

As the share of renewables increases, a structural challenge emerges: the **cannibalization effect**.

When large volumes of renewables produce simultaneously, the supply curve shifts right, depressing prices at exactly the hours of peak renewable output. This reduces the *captured price* of each additional unit of renewable capacity:

$$\text{Captured price} = \frac{\sum_h \text{Prod}_h \times \text{Price}_h}{\sum_h \text{Prod}_h}, \qquad \text{Captured ratio} = \frac{\text{Captured price}}{\overline{\text{Price}}_h}$$

Between 2021 and 2024, **solar captured value dropped by 30–40%** across major EU grids (Spain, France, Germany, Greece).

Negative Prices

Negative electricity prices are becoming frequent. In Belgium:

- **519 hours** cleared at negative prices in 2025, representing ≈6% of all hours.
- This is a more than threefold increase compared to 2024 (≈400 hours).

2.9 Flexibility: The Key Enabler of the Energy Transition

The grid must remain balanced at all times (Energy IN = Energy OUT). With rising shares of intermittent renewables, **flexibility** becomes critical.

$$\text{Flexibility} = \text{Adjusting the demand or supply curve}$$

Flexibility Strategies

1. **Local optimisation** — Control peak power (kW), reduce grid fees, manage capacity tariffs.
2. **Market optimisation** — Day-Ahead (DA) and Intraday (ID) arbitrage, imbalance (IMB) optimisation.
3. **Reserve services** — Frequency regulation (aFRR, R1/R2/R3), voltage control for the TSO.

The value of flexibility ranges from ≈5 k€/MW/year (peak management) to ≈300 k€/MW/year (primary frequency reserve R1/R2).

Battery Energy Storage Systems (BESS)

BESS is the most scalable flexibility asset:

- Global investment in BESS reached **$65.5 billion** in 2025 (source: IEA).
- In 2024, a 1–3 h BESS could earn up to **0.5 M€/MW** through combined DA/ID and ancillary market revenues.
- At current ancillary market prices, BESS projects can be profitable in **less than 5 years**.
- However, a **massive development pipeline** across EU markets (BE, DE, NL, FR, UK) risks saturating ancillary markets.

Other Flexibility Sources

- **Electric Mobility** — Smart (V1G) and bidirectional (V2G) charging.
- **Industrial demand response** — Electrolysers, process interruptions.
- **Residential** — Heat pumps, boilers, home batteries, solar curtailment.

All of these are enabled by **dynamic pricing tariffs**.

2.10 New Market Actors

The energy transition has given rise to new roles in the value chain:

- **Prosumers** — Consumers who also produce energy (e.g. households with solar panels and batteries), actively participating in both supply and demand.
- **Aggregators** — Entities that pool flexibility from many small assets and offer it to the BRP or TSO. They sign framework agreements with BRPs and flexibility contracts with prosumers.
- **Energy Service Companies (ESCOs)** — Provide energy solutions, optimisation insight, and turnkey services to industrial and commercial clients.

Chapter 3 Wind turbines

3.1 Summary: Ventis – Wind Farm of Gouy-Lez-Piétons

3.1.1 Company Overview

Ventis S.A. is a Belgian wind energy developer founded in the early 2000s by brothers Benoît and Pierre MAT, headquartered in Tournai. The company manages wind energy projects entirely in-house – from development and permitting through construction and long-term operation – operating with a local and sustainable philosophy in Wallonia and France. Its internal team covers all key disciplines: development, construction, operation, administration, and communication.

Between 2022 and 2025, Ventis installed 25% of all new wind capacity commissioned in Wallonia, recently becoming the first independent wind energy producer in the region. The company currently operates approximately 90 wind turbines across Belgium and France, representing a total installed capacity of around 250 MW, of which nearly 200 MW corresponds to its own ownership share.

In Belgium, Ventis operates wind farms across the provinces of Hainaut, Brabant wallon, and Liège, with sites in operation such as Dour–Quiévrain–Hensies (9 turbines), Tournai-Antoing-Brunehaut (10 turbines), La Louvière (5 turbines), and Quévy-Mons (7 turbines), among others. Several projects are also authorised or under development, including repowering operations and new greenfield sites. In France, Ventis operates farms in Longueville-sur-Aube (Aube, 11 turbines) and Gorges-Gonfreville (Manche, 9 turbines), with additional projects under development in Pas-de-Calais and Aisne.

3.1.2 The Gouy-Lez-Piétons Project

Permitting History

The Gouy-Lez-Piétons wind farm illustrates the complexity of the Belgian permitting process for wind energy. Ventis submitted its initial permit application in March 2014. The first decision, issued in August 2014 by the delegated and technical officers, was a refusal based on an unfavourable opinion from Belgocontrol (the Belgian air navigation service provider). Ventis appealed, obtained a conditional opinion from Belgocontrol, and was granted the permit on appeal by Minister Carlo Di Antonio in January 2015.

However, the municipality of Pont-à-Celles lodged a further appeal in March 2015, citing high landscape impact. The Council of State annulled the permit in March 2017 on the grounds that alternative locations had not been sufficiently examined. Ventis resubmitted the permit application in May 2019, providing a supplemental impact study. The permit was re-granted by the Minister in July 2019, followed by yet another appeal from the municipality in September 2019, this time contesting the examination of alternatives. Finally, in November 2022, the Council of State cleared the permit of all appeals, confirming that no alternative locations were available. From first application to final clearance, the permitting process spanned nearly nine years.

Technical Characteristics

The wind farm comprises 8 turbines arranged across agricultural land near Gouy-Lez-Piétons, connected by underground MV cables and internal access roads. The main technical characteristics are as follows:

- **Wind turbine model:** Nordex N131
- **Nominal power per turbine:** 3.6 MW
- **Total installed capacity:** 28.8 MW
- **Total height:** 150 m

- **Rotor diameter:** 131 m
- **Grid connection:** DSO ORES, 10.8 kV, 25 MW injection capacity
- **Annual energy production:** ≈57 GWh
- **Capacity factor:** ≈26%
- **Households supplied:** ≈14 400 (yearly average consumption)

Ownership Structure

Six of the eight turbines are owned by the project company GOUYNRG, while the remaining two are owned by ENGIE, which was a competing developer during the development phase. The GOUYNRG shares are distributed as follows: 70% held by Ventis, 15% by Greenplug (a development partner), and 15% by local public and citizen participation – reflecting Ventis's commitment to community involvement and local acceptability.

3.1.3 Project Development

Wind Turbine Sizing

The dimensions of the wind turbines are constrained by a combination of biological and air traffic factors. The minimum ground clearance (distance between the blade tip at its lowest point and the ground) is determined by biological constraints, particularly the presence of bat species requiring a safety margin above the tree canopy. The maximum blade tip height is set by air traffic constraints enforced by Belgocontrol. Together, these two values define the admissible rotor diameter (their difference) and, consequently, the tower height. The wind resource assessment further informs the optimal rotor size, since power extracted from the wind scales with the swept area and the cube of wind speed:

$$P_{\text{wind}} = \frac{1}{2}\rho A v^3 \quad [\text{W}]$$

where ρ is air density (approximately 1.2 kg/m^3 at 15°C), A is the rotor swept area, and v is wind speed.

Turbine Layout and Wake Effects

The number and spatial distribution of turbines is governed by the available grid injection capacity (25 MW with ORES), land acquisition constraints, local setback distances from housing, natural environments, and infrastructure, and wake effect losses. Wake effects occur when a downstream turbine operates in the turbulent, lower-energy airflow created by an upstream turbine. The prevailing wind rose at the site (predominantly south-southwest to west-southwest) directly influences the optimal turbine spacing and alignment to minimise these losses.

A constraint map overlaying setback distances from housing zones, motorways, high-voltage lines, hertzian beams, forests, and Natura 2000 areas was produced during the development phase to define the feasible turbine positions.

3.1.4 Turbine Technology

Turbine Selection Process

Three candidate turbine models were evaluated: the Vestas V126 (3.45 MW), the Nordex N131 (3.60 MW), and the Siemens Gamesa SG-132 (3.47 MW). Selection criteria included permit height limits (total height and ground clearance), rotor size, nominal power, noise output, site access feasibility, wind resource studies, and site assessment reviews.

A key finding during the site assessment was that turbine E3 (and to a lesser extent E6) faced a 15% energy curtailment constraint due to the proximity of a small woodland, reducing the net annual energy production of the SG-132 option to only 49 130 MWh/year – lower than the N131's 57 688 MWh/year. Combined with a competitive price-to-performance ratio and no turbulence-related power limitation, the Nordex N131 was selected.

Generator Architectures

Two main electrical architectures are used in the wind industry today:

Doubly-Fed Induction Generator (DFIG). Used by manufacturers such as Nordex, Vestas, and Siemens Gamesa (onshore). The rotor is connected to the grid through a partial-scale power converter (handling only 25–35% of rated power), while the stator is connected directly. This results in a smaller, cheaper converter and high efficiency at rated power. However, the direct stator–grid coupling makes the machine more sensitive to grid voltage fluctuations, and the gearbox and slip rings require more frequent maintenance.

Gearless Direct Drive (Permanent Magnet Synchronous Generator – PMSG). Used by Enercon and Siemens Gamesa (offshore). The rotor shaft drives the generator directly, eliminating the gearbox. A full-scale power converter decouples the generator entirely from the grid, enabling superior grid performance (fault ride-through, precise active and reactive power control). The absence of a gearbox reduces maintenance requirements and potentially extends the turbine's operational lifetime. The trade-off is a larger, heavier, and more expensive full-power converter chain, and more complex fault diagnostics. Enercon's design has evolved over generations: older models (e.g. E82) placed the rectifier in the nacelle and the inverter at the tower base via a DC link, while current models (E115) integrate the full conversion chain in the nacelle, and newer large models (E175) consolidate all power electronics in a compact nacelle structure.

› 3.1.5 Construction

Logistics and Transport

Wind turbine transportation is one of the most challenging aspects of onshore wind projects in densely populated regions such as Belgium. Components – blades (approximately 65 m), tower sections, and nacelles (approximately 13 m long, 4 m wide) – are manufactured in Europe and transported by sea to the port of Antwerp. A local specialised transport company then takes over responsibility for the convoy from the port to the site, including liability for damage or blockages.

The transport company conducts detailed route surveys, produces turn simulations using dedicated software, and performs dummy runs (typically at night) to validate the route before actual component delivery. Road adaptations may include removing road signs, applying steel plates on roundabout islands, and using high-density polyethylene (PEHD) ground protection plates to extend turning radii on agricultural land. From the road transition point (RTP) onwards, Ventis assumes full responsibility for road design, land agreements, and any delays or damage.

Civil Works (Balance of Plant)

Access roads and crane platforms must meet strict bearing capacity requirements: up to 12 t per axle and a total crane load of up to 137 t. Road layers are designed using finite element modelling (FEM) based on geotechnical reports and OEM specifications, and validated on-site by plate load tests (EV1/EV2 compaction ratios). Water management (drainage, avoiding mud and icing) is also a critical design criterion.

Foundations. Two main categories of foundation design are used: slab (or gravity) foundations and pile foundations. The choice depends on geological conditions (pile foundations are unsuitable for karstic limestone), available land area, groundwater level (foundations are elevated to reduce buoyancy and increase hub height), and cost (depth to competent rock is a key driver). A geotechnical campaign – comprising penetration tests, core drilling, electrical resistivity tomography, and piezometer installation – is mandatory to characterise the subsoil. The Gouy-Lez-Piétons foundations have a diameter of 19.2 m, compared to 17.6 m for a comparable 2 MW turbine from 2007, reflecting the increased loads of modern larger machines.

Gravity foundation variants include direct soil replacement and soil improvement techniques (ballasted columns, rigid inclusions, or mixed inclusions). Pile foundations may use driven piles (steel or concrete, impact-driven) or bored piles (various auger techniques). Concreting of the slab is performed in a single continuous pour, and the foundation is then backfilled and compacted.

Electrical Infrastructure

MV cables (24 kV, aluminium, cross-sections of 150 or 240 mm^2) are buried at a depth of approximately 1.3 m, laid in a bed of sand with PE protection slabs and warning ribbons, alongside HDPE conduits for fibre-optic communication cables.

The on-site electrical cabin contains two distinct sections. The high-voltage (HV) side houses ten SF6 switchgear cells: three for ENGIE (departure, metering, generator protection), four for ORES (two arrival feeders, reference voltage, ORES protection), and three for Ventis/GOUYNRG (generator protection, metering, departure), plus an auxiliary transformer. The low-voltage (LV) side contains the SCADA controllers for each owner (Nordex

CWE units), an aggregator box for curtailment management based on electricity prices, a NORTEC unit for automated shadow flicker and bat curtailment, a Phoenix UPS, internet routers with fixed IPs and firewalls, and a 4G backup system.

Turbine Assembly and Commissioning

Assembly involves the positioning of a main crawler crane (LTR1100) and an assist crane (LTM1650-8.1) according to OEM lifting plans. Tower sections, nacelle, hub, and blades are lifted and assembled sequentially. Commissioning includes electrical completion, fibre-optic cabling, functional test runs, grid energisation, and formal inspections by the DSO and a certified independent body (e.g. TÜV, Vinçotte, Bureau Veritas). A punch list process formalises the handover between the OEM and the owner.

3.1.6 Operation and Maintenance

Operation of the wind farm encompasses preventive maintenance, corrective maintenance, remote monitoring, mandatory regulatory inspections, and environmental monitoring.

Preventive maintenance includes periodic bolt re-torquing (tower flanges, nacelle, blade roots), regreasing of gearboxes, pitch bearings, and yaw rings, visual inspections of blades, nacelle components and tower internals, functional testing of safety systems, and planned replacement of wear parts. A condition monitoring system (CMS) using vibration sensors on the main bearing and gearbox allows early detection of anomalies such as under-greased bearings, enabling intervention before catastrophic failure.

Corrective maintenance covers unplanned repairs such as blade damage from lightning strikes (repaired by rope-access technicians) and major component replacements such as generator swaps, which require a dedicated service crane.

Remote monitoring is performed via the Farsight SCADA platform, which provides real-time dashboards, power curve analysis, wind analysis, alarm management, and reporting for each turbine across the portfolio.

Environmental monitoring obligations include post-construction bat and bird mortality surveys (using trained dogs and regular transect walks), shadow flicker management (automated curtailment via NORTEC when a receptor exceeds the permitted annual shadow hours), noise measurements at neighbouring dwellings, and ice detection on blades.

3.1.7 Grid Constraints and Power Offtake Optimisation

Curtailments

The 25 MW injection capacity agreed with ORES limits the farm's ability to export its full 28.8 MW installed capacity at all times. Additionally, curtailments are triggered when day-ahead or intraday electricity market prices turn negative, since injecting power at negative prices generates a direct financial loss. Environmental curtailments (bats, shadow, noise) add further stops, typically representing 4% of gross annual production at Gouy.

Grid Congestion Context

The Belgian distribution grid is under significant stress: renewable energy connection requests to ORES increased by more than a factor of four between recent years, and the total demand connected or pending connection is growing rapidly (from approximately 350 MW measured load to projections above 500 MW by 2025), driven by battery storage, EV fast charging, and industrial electrification. The grid is aging and was designed for a unidirectional power flow from large central plants to consumers – the opposite of distributed renewable generation.

3.1.8 Future Developments

Ventis identifies four main optimisation vectors to improve the economic viability of its portfolio under increasing grid congestion and market volatility:

Behind-the-meter direct connection (private PPA). A direct electrical line between a wind farm and a nearby industrial customer bypasses the distribution grid entirely, eliminating network tariffs. Both parties benefit: the producer receives a higher price than the spot market, and the industrial customer pays less than the regulated tariff. This arrangement also tends to facilitate permitting, as industrials committing to green energy consumption are viewed favourably. Battery storage can further improve the self-consumption rate.

Repowering. End-of-life turbines are replaced by fewer but larger and more powerful machines, making use of the existing grid connection, partially secured land, and an already-integrated landscape. Modern turbines deliver significantly more energy per unit cost than those installed 20 years ago. Permitting is generally more straightforward, as the visual impact on the landscape is already accepted by the local community and the turbine count decreases.

Battery Energy Storage Systems (BESS). BESS co-located with wind farms can absorb energy that would otherwise be curtailed (due to grid limits or negative prices) and re-inject it later. They also enable participation in ancillary service markets (frequency containment reserve FCR, automatic frequency restoration reserve aFRR, imbalance settlement) and support peak-shaving strategies. One concrete project under development involves a 2 MW / 4 MWh BESS at a 10.8 MW wind farm (PEDG) with an 8.8 MW injection limit.

EV fast-charging stations. Wind turbines can supply EV chargers through a direct private line, providing traceable renewable electricity for mobility without loading the public grid. This creates a win-win situation for the producer (better selling price) and the consumer (cheaper green electricity), reduces the need for municipalities to fund public charging infrastructure, and increases local acceptance of associated storage or wind projects. A concept project (MWPk) envisions approximately 10 fast chargers near a motorway junction with a potential of 50 000 vehicles per day, powered by a 9.4 MW wind farm with no impact on the local distribution grid.

Wind/solar hybridisation. Solar PV generation is strongly complementary to wind in Belgium: wind output peaks in winter and at night, while solar generation peaks in summer and midday. Combining both sources at a shared grid connection point improves the utilisation of the grid capacity and the associated BESS, reduces the net cost of grid connection for the solar project, and creates opportunities for agrivoltaic development beneficial to local farmers. A project under study (EAE) combines a 25 MW wind farm with a 700 kWp solar installation and a 3.5 MW / 7 MWh BESS.

Chapter 4 Slow Heat

4.1 SlowHEAT – Heat the Body, Not the Building

Synthesis of the presentation by Prof. Geoffrey van Moeseke, UCLouvain, February 18, 2026

4.1.1 Core Intuition and Motivation

The buildings sector is one of the largest contributors to greenhouse gas emissions in Europe, and space heating accounts for the bulk of residential energy consumption. The dominant policy response has been to invest massively in building renovation — improving insulation, replacing windows, and upgrading heating systems. The EU renovation wave is estimated to require roughly €243 billion per year to achieve its climate objectives. Yet a simple thought experiment reveals the staggering scale of this commitment: equipping every EU citizen with high-quality insulating outerwear would cost approximately €12.5 billion per year — nearly twenty times less.

This provocative comparison is not a call to replace renovation with ski suits, but rather an invitation to question a deeply embedded assumption: that thermal comfort necessarily means heating the entire volume of a building to a uniform temperature. The SlowHEAT project, led by Prof. Geoffrey van Moeseke at UCLouvain and conducted in collaboration with UCLouvain's iPSY institute, the *habitat et participation* association, LAB Architecture, *Architecture et Climat*, and ULB-IGEAT, with support from Innoviris Brussels, explores a radically different paradigm. Rather than heating the building, the core idea is to direct heat toward the occupant's body through *Personal Comfort Systems* (PCS) — a concept with deep historical roots and a growing body of scientific support.

4.1.2 An Idea Rooted in History

The notion of localised, body-centred heating is far from new. For millennia, heat was confined to specific spots within the dwelling. Ancient Greek braziers, Gaulish central hearths, Viking longhouse fire pits — all provided radiant warmth to those gathered around them, while leaving the rest of the space cold. Furniture was similarly designed to concentrate warmth: canopied and curtained beds created insulated sleeping micro-environments, hooded chairs shielded occupants from draughts, folding screens blocked cold air, and small portable foot warmers carried embers beneath one's skirts. Japanese kotatsu tables, combining a low table with a heating element and a heavy draping blanket, are a particularly elegant example of the same principle: heat the body, not the room.

The shift toward uniform centralised heating began in earnest in the late nineteenth and early twentieth centuries, with the industrialisation of cast-iron radiators and gas or coal boilers. This transition was not purely technological — it was also cultural. Advertising campaigns of the 1930s (including those of the Compagnie Nationale des Radiateurs in France and the National Radiator Company in England) actively constructed a new narrative: central heating was modern, hygienic, uniform, effortless, and liberating. The old way of warming — gathering around a stove, layering clothing indoors, retreating to a heated corner of the house — was reframed as backward, uncomfortable, and redolent of poverty.

The consequences have been significant. Historical data show that average indoor winter temperatures in UK homes rose from approximately 15–16 °C in the 1960s to 19–20 °C by the late 1990s. Research by Mavrogianni et al. (2013) documents this steady upward drift across living rooms, bedrooms, and hallways alike. At the EU scale, the European Environment Agency has estimated that switching from single-room heating to central heating increases energy demand for space heating by roughly 25 % on average — an increase that has partly offset the efficiency gains achieved through better insulation and more efficient boilers over the same period.

4.1.3 An Idea Aiming for Progress

Far from being a nostalgic regression, the SlowHEAT approach is positioned as a forward-looking response to multiple contemporary challenges.

Improved health. Recent work published in *Energy & Buildings* (Zhu et al., 2025) underscores that prolonged exposure to thermoneutral environments may compromise both comfort and health. The human body is equipped with a sophisticated array of thermoregulatory mechanisms — vasoconstriction and vasodilation, piloerection, brown adipose tissue activation, and shivering thermogenesis — that are simply not engaged when indoor temperatures are maintained at a constant 20–22 °C. Mild, repeated cold exposure has been associated with metabolic benefits, including activation of brown fat, as demonstrated by van der Lans et al. (2013). The hormesis principle — whereby moderate physiological stress produces adaptive benefits — may apply to thermal environments just as it does to exercise.

Improved individual wellbeing. Unlike conventional HVAC systems that impose a single thermal setpoint on all occupants of a shared space, PCS allow individuals to modulate their own thermal environment without affecting those around them. Research on office environments (Wegertseder-Martinez et al.) confirms that this individualised control improves both comfort satisfaction and perceived wellbeing.

A pathway toward electrification. The phase-out of fossil fuel heating is a central objective of European climate policy. In Belgium, oil boilers are already banned in new constructions in Wallonia (from January 2026), and Brussels has extended the ban to gas boilers as well. Heat pumps are the primary proposed alternative, but their high upfront cost and the need for low-temperature distribution systems make them challenging to retrofit. Low-power electric PCS — consuming tens to a few hundred watts rather than several kilowatts — represent a more immediately accessible bridge technology, compatible with existing electrical infrastructure.

Technological and social pluralism. The approach does not prescribe a single solution. It accommodates both high-tech options (battery-powered heated garments, smart infrared panels, app-controlled electric blankets) and low-tech ones (well-chosen woollen underlayers, homemade heated capes fabricated in community repair workshops, draught exclusion). This breadth means it can appeal to both techno-optimists and advocates of degrowth and sufficiency.

4.1.4 The SlowHEAT Living-Lab Experiment (2020–2023)

General Framework

The empirical core of SlowHEAT is a three-year living-lab study conducted with thirty volunteer households in Belgium, running from winter 2020–2021 through winter 2022–2023. The study period coincided, unplanned, with the 2021–2022 energy price crisis triggered by the Russian invasion of Ukraine — an external shock that adds both complexity and ecological validity to the results.

Participants were voluntary, self-selected, and environmentally motivated — a sample the authors themselves acknowledge as likely to produce over-optimistic results compared to the general population. The group was predominantly highly educated (64 % held a master's degree, 14 % a PhD), with a balanced gender distribution and a spread of age groups from the 20s to the 60s, with the largest cohort in their 40s (31 %). Housing types and energy performance levels were deliberately varied.

The methodological framework drew on social practice theory, following Schatzki (1997) and Gram-Hanssen (2010), analysing heating behaviour along four interacting dimensions: material infrastructures and objects, practical knowledge and skills, representations and meanings, and collective norms and rules. Participants were not asked to endure discomfort: the goal was to find a personal equilibrium, supported by a toolkit of PCS made available to all households. These ranged from heated bed covers (100 W) and heating chair pads (30 W) to small infrared panels (300 W), radiant heaters (1 200 W), large infrared panels (400 W), and heated desk pads (80 W).

Quantitative Results

The central quantitative findings are striking. Over the course of the project, participants progressively reduced their indoor temperatures well below conventional norms. Table 4.1 summarises the evolution of living-room temperatures.

In terms of energy consumption, participants achieved average reductions of approximately 50 % in gas consumption on a heating-degree-day adjusted basis, and approximately 14 % in electricity consumption. The authors note two important caveats: gas savings may be overestimated because most dwellings are apartments

Table 4.1: Indoor living-room temperature: thermostat setpoints before the project and measured temperatures during the project. Medians, means, and standard deviations are calculated for the whole participant group. Both occupied and unoccupied periods are included.

Period	Metric	Median (°C)	Mean (°C)	SD (°C)
Before project	Thermostat setting	19.0	19.0	1.6
First winter	Coldest week	16.7	17.6	1.9
	Coldest day	15.8	16.9	2.1
Third winter	Coldest two months	15.0	15.1	1.9
	Coldest month	14.5	14.7	2.0
	Coldest week	13.7	13.5	2.6
	Coldest day	12.5	12.6	2.5

or semi-detached houses that benefit from heat transfers from neighbouring units; and the electricity savings were unexpected, likely reflecting the broader behavioural response to the 2022 energy price shock rather than the PCS alone.

Inter-household variation was considerable. Mean indoor temperatures across participating dwellings ranged from approximately 11 °C to 16 °C, reflecting wide differences in housing performance, household composition, and personal thermal preferences. One particularly illustrative case is that of a household that had already undergone a building renovation: over the same period, SlowHEAT practices generated savings of 700 m^3 of gas per year, compared to 380 m^3 per year attributable to the renovation itself.

4.1.5 Qualitative Analysis: A Social-Practice Framework

Changing Representations

Perhaps the deepest shift required by the SlowHEAT approach is representational. Our current expectation that a home should be uniformly heated to 19–22 °C is not a biological necessity but a historically and socially constructed norm — one that, as historian Olivier Jandot argues, is inseparable from the technological evolution of the nineteenth and twentieth centuries. This norm is recent, geographically variable, and therefore changeable.

One of the most striking indicators of this normative rigidity is the public reaction to media coverage of the SlowHEAT project. Social media comments cited in the presentation include accusations of endangering children, comparisons to medieval living conditions, and assertions that living below 20 °C is simply unacceptable in 2022. These reactions illustrate the depth of the cultural embedding of centralised warmth as a marker of dignity and modernity — an embedding that was itself deliberately constructed by the heating industry a century ago.

Alongside this cultural shift, participants were invited to develop a richer vocabulary for bodily thermal sensations and to develop awareness of their body's own thermoregulatory capacities: vasoconstriction, brown adipose tissue activation, and the adaptive reduction in perceived cold discomfort with regular mild exposure.

Acquiring Knowledge and Skills

A significant part of the learning process concerned the relationship between indoor temperature, relative humidity, and air quality. A key practical insight is that, at 15 °C, maintaining indoor relative humidity below approximately 55 % (corresponding to a humidity ratio of roughly 0.006 g_{water}/kg_{air}) is sufficient to prevent mould growth on most building materials. Critically, in a Belgian winter climate, outdoor air humidity rarely exceeds this threshold, meaning that regular ventilation with outdoor air is almost always safe from a mould-prevention perspective — a counterintuitive finding for many participants who feared that cooler interiors would be moister and more prone to mould.

The SlowHEAT team also developed a dedicated tool, the *SlowZone* widget, which combines real-time outdoor temperature and humidity data with psychrometric calculations to indicate the safe ventilation window for any given location.

Participants also developed practical skills in selecting and deploying PCS: understanding the difference between conductive heating (body contact), radiant heating (infrared proximity), and convective heating (air warming), and learning which solution best matched each activity, body zone, and time of day.

Infrastructure and Technology: A Hierarchy of Solutions

The project proposes a structured hierarchy of heating interventions, ordered from least to most energy-intensive, as summarised in Table 4.2.

Table 4.2: Proposed hierarchy of Personal Comfort System interventions, from zero-energy measures to whole-building heating.

Class	Power	Solution family	Examples
A – Zero-energy measures	0 W	Clothing, zoning, activity	Thermal underwear, closing doors, physical activity, curtains
B – Body-contact conduction	±50 W/person	Wearable or contact accessories	Heated chair pad, electric blanket, heated cape, hot-water bottle
C – Radiant proximity	±300 W/person	Radiant or heated furniture near occupant	Infrared panel, radiant column, heated table
D – Local room heating	±1 500 W/room	Thermostatic room heater	Room maintained at 15–17 °C when occupied, with Classes A–C still active
E – Whole-building heating	±5 000 W/dwelling	Central heating system	Kept at frost-protection level (8 °C) or passage temperature (12–15 °C)

The philosophy is not to eliminate central heating entirely, but to relegate it to a supporting, background role while Classes A through C meet most comfort needs. Low-tech Class A and B solutions are emphasised: the project ran community sewing workshops to produce washable heated capes equipped with recycled heating textile, and explored prototype heated chair inserts and desk setups using low-voltage heating elements.

Navigating Social Norms

The social dimension proved one of the most challenging aspects of the experiment. Participants navigated real tensions between their new practices and the expectations of visitors, family members, and social convention. Receiving guests in a 15 °C home, asking in-laws to put on a jumper, or negotiating thermostat settings within a household all require a form of social courage and communication that purely technological solutions do not demand.

Some participants experienced these tensions positively, as a sign of being ahead of their time. Others found the social cost significant. The project also surfaced a regulatory tension: current energy performance certificate frameworks in Wallonia (and more broadly across Europe) rate buildings on the basis of normalised energy consumption at standard occupancy and temperature assumptions. A household that genuinely heats its home to 14 °C and achieves excellent comfort through PCS may appear, on paper, to have a poorly performing building — or may even find that certain renovation subsidies are inaccessible because its actual consumption already falls below the required thresholds. The authors argue that regulation should distinguish more clearly between the *means* (energy in kWh/m^2) and the *end* (a healthy, enabling domestic environment), and that "off-framework" renovation pathways tailored to sober lifestyles should be legally recognised.

4.1.6 Discussion: Efficiency, Sufficiency, and Beyond

SlowHEAT raises fundamental questions about the relationship between efficiency and sufficiency in energy policy. The dominant paradigm — enshrined in frameworks such as the Negawatt approach — places sufficiency first, followed by efficiency, then renewable energy. In practice, however, sufficiency is rarely operationalised with the same rigour as efficiency: it tends to mean marginal reductions (lower the thermostat by 1 °C, reduce shower time by two minutes) rather than a qualitative transformation of how services are delivered.

SlowHEAT illustrates what a more radical form of sufficiency can look like in practice: not "less of the same" (a slightly cooler uniformly heated home) but a fundamentally different service configuration (a cool home with locally warm people). This distinction is important not only for its energy implications but for its social ones. As the concept of *consumption corridors* (Akenji et al., 2021) suggests, sufficiency should be bounded both by a social floor — below which underconsumption causes deprivation and health risk — and an environmental ceiling — above which overconsumption is ecologically unsustainable. Both boundaries should be democratically established and informed by science. The SlowHEAT experiment suggests that the social floor for thermal comfort may be considerably lower than current norms imply, provided adequate PCS are available and occupants have the knowledge, agency, and social support to use them.

The project's main acknowledged limitations are twofold. First, the sample's high educational level and prior environmental motivation make it difficult to generalise: a participant group with mixed motivations and lower climate awareness might achieve smaller reductions and report lower satisfaction. Second, the energy savings attributed to SlowHEAT may be partially overestimated in multi-unit dwellings, where heat transfers from neighbouring heated apartments contribute to maintaining indoor temperatures. More controlled studies, with matched comparison groups and sub-metered heating data, are needed to isolate the specific contribution of PCS from confounding behavioural and contextual factors.

Notwithstanding these limitations, the SlowHEAT project makes a compelling empirical case that a significant fraction of European households could, with appropriate support, substantially reduce their heating energy consumption while maintaining subjective comfort — and that the barriers to doing so are as much cultural, regulatory, and social as they are technical.

Chapter 5 Batteries

5.1 Introduction: Energy Storage and Electrochemistry

The growing demand for portable energy, electric vehicles, and renewable energy integration has made efficient energy storage a central challenge of our time. Raw energy — whether from fossil fuels, nuclear reactions, solar irradiation, or wind — must often be transformed and stored before use. Among storage technologies, electrochemical batteries offer a unique combination of portability, scalability, and high energy conversion efficiency.

The classical thermomechanical route to electricity — combustion, thermal engine, alternator — is fundamentally limited by the Carnot principle, which caps efficiency at $\eta_{\max} = 1 - T_c/T_h$. The electrochemical route, by contrast, achieves a direct conversion of chemical energy into electrical work, with a theoretical yield given by:

$$\eta_{\text{th}} = \frac{\Delta G}{\Delta H} \tag{5.1}$$

For a hydrogen/oxygen fuel cell, this yields $\eta_{\max} \approx 95\%$, far exceeding any heat engine operating between practical temperature limits.

The fundamental unit of an electrochemical energy storage system is the **electrochemical cell**, composed of two electrodes separated by an electrolyte:

electrode / electrolyte / electrode

At constant temperature and pressure, the Gibbs free energy of the cell reaction is directly related to the cell voltage:

$$\Delta G(T, P) = -nF\Delta\varepsilon \tag{5.2}$$

where n is the number of electrons exchanged, $F = 96{,}485\ \text{C/mol}$ is Faraday's constant, and $\Delta\varepsilon$ is the electromotive force. The spontaneity condition requires $\Delta\varepsilon > 0$. The maximum achievable cell voltage is:

$$\Delta V_{\max} = \Delta\varepsilon = -\frac{\Delta G}{nF} \tag{5.3}$$

For example, the lead-acid cell gives $\Delta V_{\max} = 2.1\ \text{V}$ per cell, so a standard 12 V car battery consists of six cells in series.

Rechargeable (secondary) batteries must additionally satisfy three conditions: the cell reaction must be reversible, the charge/discharge cycling yield must be acceptable, and material degradation must remain within tolerable bounds.

The history of rechargeable batteries spans nearly two centuries, from Planté's lead-acid cell (1859) through nickel-cadmium and nickel-metal hydride technologies, culminating in the lithium-ion battery developed between 1970 and 1990 by Whittingham, Goodenough, and Yoshino — work recognised by the 2019 Nobel Prize in Chemistry. Over this period, the specific energy of rechargeable batteries has risen from roughly 30 Wh/kg (lead-acid) to over 180 Wh/kg (Li-ion), with volumetric energy density approaching the theoretical limit of around 400 Wh/L.

5.2 How a Li-ion Battery Works

5.2.1 General Operating Principle

A lithium-ion battery operates through the **reversible intercalation** of lithium ions (Li^+) between two host electrode materials. Crucially, lithium is never reduced to its metallic form during normal operation; it remains ionic throughout the charge/discharge cycle.

- **Negative electrode (anode):** graphite, which hosts Li^+ ions in its layered structure to form LiC_6.
- **Positive electrode (cathode):** a lithiated transition metal oxide, typically of the form $LiMO_x$ (e.g., $LiCoO_2$, $LiFePO_4$, $LiMn_2O_4$, or NMC variants).

The overall cell reaction can be written as:

$$\mathrm{LiMO}_x + \mathrm{C}_n \underset{\text{charge}}{\overset{\text{discharge}}{\rightleftharpoons}} \mathrm{Li}_{1-y}\mathrm{MO}_x + \mathrm{Li}_y\mathrm{C}_n \tag{5.4}$$

During **discharge**, Li^+ ions de-intercalate from the graphite anode, travel through the electrolyte, and re-intercalate into the cathode oxide, while electrons flow through the external circuit producing electrical work. During **charge**, the process is reversed.

5.2.2 Criteria for Electrode Materials

An effective intercalation electrode material must satisfy several criteria simultaneously:

1. **High Li^+ insertion capacity:** the material must accommodate a large number of lithium ions per formula unit, yielding high specific capacity (mAh/g).
2. **Reversibility:** insertion and de-insertion must be highly reversible with minimal hysteresis, ensuring long cycle life.
3. **Voltage plateau:** the electrode potential should be stable (flat plateau) at an adequate value with respect to Li^+/Li.
4. **Good electronic and ionic conductivity:** both electron and ion transport through the electrode must be fast enough to support practical current densities.
5. **Low volume change:** the crystal structure should expand and contract minimally upon Li^+ insertion/extraction to avoid mechanical degradation.
6. **Durability and low cost:** the material must be stable over hundreds to thousands of cycles and manufacturable from abundant, inexpensive precursors.

The stored energy of a cell is given by:

$$E = \Delta U \times i \times t \tag{5.5}$$

where ΔU is the electromotive force (difference between cathode and anode potentials) and $i \times t$ is the capacity delivered.

5.2.3 Representative Electrode Materials

Table 5.1 summarises the key properties of common positive and negative electrode materials.

Li-ion cells typically deliver 3.5–3.8 V, compared to 2.0 V for lead-acid and 1.2 V for Ni-MH cells. Compared to competing rechargeable technologies, Li-ion offers a superior energy density (180–220 Wh/kg vs. 30–80 Wh/kg for older chemistries), faster charging (60–120 min), lower self-discharge (5–10%/month), and a charge/discharge round-trip efficiency exceeding 85%.

5.3 Battery Construction

5.3.1 Cell Architecture

The electrochemical cell consists of thin-film layers stacked or wound together:

current collector | anode | electrolyte | separator | electrolyte | cathode | current collector

Table 5.1: Properties of selected Li-ion electrode materials.

Material	Avg. voltage	Specific capacity	Mass energy
Positive electrodes			
$LiCoO_2$	3.7 V	140 mAh/g	0.518 kWh/kg
$LiMn_2O_4$	4.0 V	100 mAh/g	0.400 kWh/kg
$LiFePO_4$	3.3 V	150 mAh/g	0.495 kWh/kg
Negative electrodes			
Graphite (LiC_6)	0.1–0.2 V	372 mAh/g	0.037–0.074 kWh/kg
Titanate ($Li_4Ti_5O_{12}$)	1–2 V	160 mAh/g	0.16–0.32 kWh/kg
Si ($Li_{4.4}Si$)	0.5–1 V	4212 mAh/g	2.1–4.2 kWh/kg

The anode current collector is copper foil; the cathode current collector is aluminium foil. The separator is a porous polymer membrane that prevents electronic contact between the two electrodes while allowing ionic transport.

5.3.2 Composite Electrodes

Practical electrodes are not monolithic but **composite structures** deposited as thin films (typically 100 µm thick) on the metallic current collector. Each electrode layer contains:

- **Active material** (powder): the intercalation compound itself.
- **Binder**: typically polyvinylidene fluoride (PVDF), dissolved in N-methylpyrrolidone (NMP) to form a slurry. PVDF is chemically stable but is a fluorinated polymer and therefore environmentally persistent and difficult to recycle. Alternative water-based binders (cellulose-derived gums, carboxymethyl cellulose) are actively being developed.
- **Conductive additive**: usually carbon black, to ensure electronic percolation through the electrode since many active materials have low intrinsic electronic conductivity.

The assembled electrode is soaked with liquid electrolyte, which fills the porous space between particles and ensures ionic access to all active surfaces. Nanostructuring the active material (reducing particle size, introducing porosity) is a key strategy to maximise accessible surface area and reduce solid-state diffusion distances for Li^+.

5.3.3 Electrolyte

The electrolyte must be a pure ionic conductor (electron insulator) that is stable across the full voltage window of the cell (∼4 V for standard Li-ion). Two main types are used:

- **Liquid electrolyte** ("classical" Li-ion): a lithium salt (typically $LiPF_6$) dissolved in a mixture of organic carbonate solvents (e.g., ethylene carbonate, dimethyl carbonate). These electrolytes are highly flammable, requiring the addition of flame-retardant additives and imposing strict purity requirements during manufacture.
- **Gel polymer electrolyte** (Li-ion "polymer"): a polymer matrix swollen with liquid electrolyte, offering improved safety and mechanical flexibility.

A critical limitation of liquid electrolytes is that their ionic conductivity decreases significantly at low temperatures, degrading battery performance in cold climates.

5.3.4 Cell Formats and Assembly

Because a single cell provides only 3.5–3.8 V, practical battery packs assemble many cells in series (to increase voltage) and in parallel (to increase capacity). The cell format can be cylindrical (wound jelly-roll), prismatic (stacked layers), or pouch (flat flexible). Battery manufacturing is a complex multi-step process involving slurry mixing, coating, drying, calendering, slitting, winding or stacking, electrolyte filling, and final sealing, all under tightly controlled atmospheric conditions to exclude moisture and metallic contaminants. Even microscopic metal particles can cause internal short circuits, thermal runaway, and fire.

At the pack level, a **Battery Management System (BMS)** is essential to monitor cell voltages, temperatures, and state of charge, and to balance cells, prevent overcharge/overdischarge, and manage thermal conditions.

5.4 Challenges

5.4.1 Materials Challenges

The development of Li-ion batteries relies on several materials that are classified as **critical raw materials** under EU legislation, including lithium, cobalt, nickel, manganese, and natural graphite. These elements are geographically concentrated, subject to supply chain risks, and often associated with environmental and ethical concerns in mining. This drives intensive research into alternative chemistries.

Silicon is a particularly attractive anode candidate, with a theoretical capacity of $\sim$4000 mAh/g compared to only 372 mAh/g for graphite. However, Si undergoes a volume expansion of $\sim$320% upon full lithiation (to $Li_{22}Si_5$), causing mechanical cracking and loss of electrical contact upon cycling. Strategies to mitigate this include nanostructured silicon (nanoparticles, nanowires, honeycombs, micropillars), porous composite architectures embedding Si nanoparticles in a compliant carbon matrix, and silicon-graphite blended anodes already entering commercial production.

On the cathode side, cobalt-free or cobalt-reduced NMC ($LiNi_xMn_yCo_zO_2$) formulations with high nickel content (NMC811, NMC90) are being pursued to reduce cost and supply risk while improving energy density. $LiFePO_4$ (LFP) has re-emerged strongly as a safe, low-cost, cobalt-free cathode with excellent cycle life, particularly for stationary storage and lower-range EVs.

5.4.2 Manufacturing and Component Challenges

Component-level challenges include improving electrode adhesion, electronic conductivity, and electrolyte compatibility. The replacement of PVDF/NMP binder systems with water-based alternatives (biopolymer gums such as xanthan or carboxymethylcellulose) is an active area, motivated both by environmental concerns and by the need to reduce production costs and eliminate solvent recovery steps.

At the battery assembly level, the absence of any metallic contamination is a paramount constraint, since even sub-millimetre metal particles can provoke internal short circuits leading to thermal runaway. Clean-room manufacturing environments, continuous monitoring, and automated optical inspection are indispensable.

5.4.3 Cost Trajectory

The price of Li-ion battery packs has fallen dramatically: from approximately \$780/kWh in 2013 to around \$139/kWh in 2023. The widely cited cost target of \$100/kWh is considered the threshold at which electric vehicles reach purchase price parity with internal combustion vehicles without subsidies. Continued cost reduction requires simultaneously improving energy density, reducing reliance on expensive materials, and scaling manufacturing.

5.4.4 Safety, Recycling, and End-of-Life

Safety remains a concern due to the flammability of liquid electrolytes and the risk of thermal runaway. Solid-state electrolytes — ceramics, sulfides, or polymer-ceramic composites — promise intrinsically safer batteries by eliminating flammable liquids and enabling lithium metal anodes, but achieving sufficient ionic conductivity and stable electrode/electrolyte interfaces at room temperature remains an unsolved challenge.

Recycling is increasingly critical both for resource recovery and for reducing the environmental footprint of battery production. Current recycling routes (pyrometallurgy, hydrometallurgy, direct recycling) recover metals to varying degrees but remain costly and energy-intensive. The EU Battery Regulation introduces mandatory recycled content targets and collection obligations.

5.5 Future Perspectives

5.5.1 Beyond Li-ion: Sodium-Ion Batteries

Sodium-ion (Na-ion) batteries follow the same intercalation principle as Li-ion but replace Li^+ with Na^+. Sodium is vastly more abundant and geographically distributed than lithium, and Na-ion cells can be assembled without copper current collectors (since sodium does not alloy with aluminium at low potentials, aluminium can be used for both electrodes). Water-based electrode manufacturing is also feasible. Current Na-ion cells reach around 100 Wh/kg with cycle lives exceeding 2000 cycles — lower than state-of-the-art Li-ion but improving rapidly. Cathode materials include sodium vanadium fluorophosphates ($NaVPO_4F$ and variants); hard carbon is the preferred anode. Industry is moving fast: BYD announced in February 2026 a third-generation Na-ion pack targeting 10,000 cycles, alongside a solid-state battery programme for 2027 EVs.

5.5.2 Solid-State Batteries

Solid-state batteries replace the liquid electrolyte with a solid ionic conductor (oxide, sulfide, or polymer-based). They promise higher energy density (by enabling lithium metal anodes), intrinsic safety (no flammable liquid), and potentially wider operating temperature ranges. Key unresolved challenges include: achieving room-temperature ionic conductivity comparable to liquid electrolytes ($\sim 10^{-2}$ S/cm), ensuring mechanically stable and low-resistance electrode/electrolyte interfaces over thousands of cycles, and scaling up manufacturing.

5.5.3 A Broader Perspective

Battery technology is ultimately only one part of the energy transition. Research cited in the presentation (Barrett et al., *Nature Energy*, 2022) estimates that current per-capita energy consumption in developed countries averages $\sim$6 kW/person, while sustainability analyses suggest a target of $\sim$2 kW/person is necessary. Improving energy storage is therefore inseparable from the broader imperative to reduce absolute energy demand.

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Chapter 7 Fluxys

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